2Market Retail data analysis

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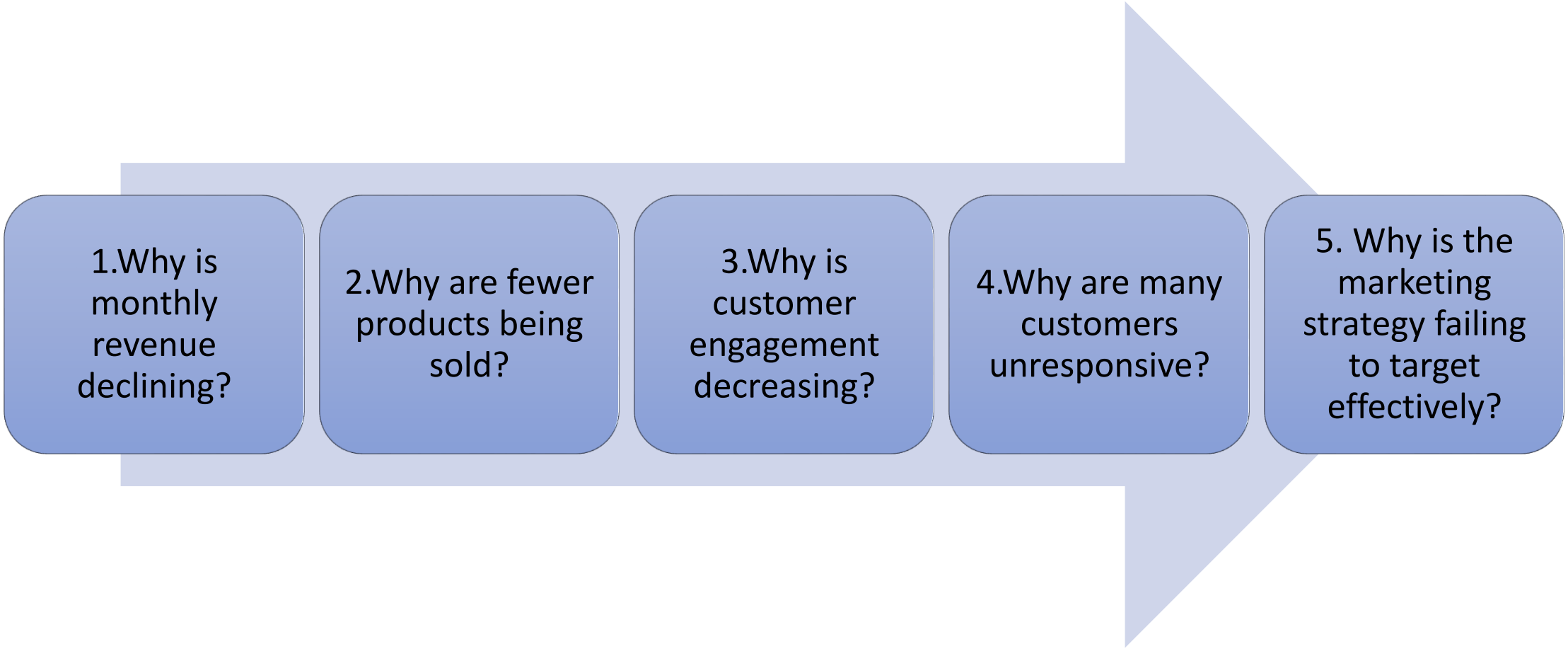
Problem statement:

- Total sales are on the declining: average revenue decreases approximately 865 units per month. This decline is due to two key issues:
- (1) significant month-to-month sales driven by seasonality and (2) limited and uneven customer engagement.
- Customers responding to multiple campaigns generate significantly higher average revenue, however, the majority of customers remain unresponsive, limiting overall sales growth.
- Customer response data reveals peak effectiveness at single ad exposures, diminishing returns and possible ad fatigue as exposure increases.
- These factors indicate the current marketing strategy struggles to expand its engaged customer base and fails to fully capitalise on high-value segments, limiting revenue growth.

5 whys


1.Why is monthly revenue declining?
2.Why are fewer products being sold?
3.Why is customer engagement decreasing?
4.Why are many customers unresponsive?
5. Why is the marketing strategy failing to target effectively?

Customer demography

Age

Range from 29 to 132 years, median of 55 and an average of age of 56 years.

- Highest proportion of 50–59: 30.6%
- About 50% of customers are between 48-66 yrs.

Income

Ranges from 1730 to 666666, a median income of 51382 and an average of about 52247

About 50% of customers earn between 35303 -68522 (25-75th percentile).

- Lower-Mid(30,000 – 49,999): 31.14%

Education levels

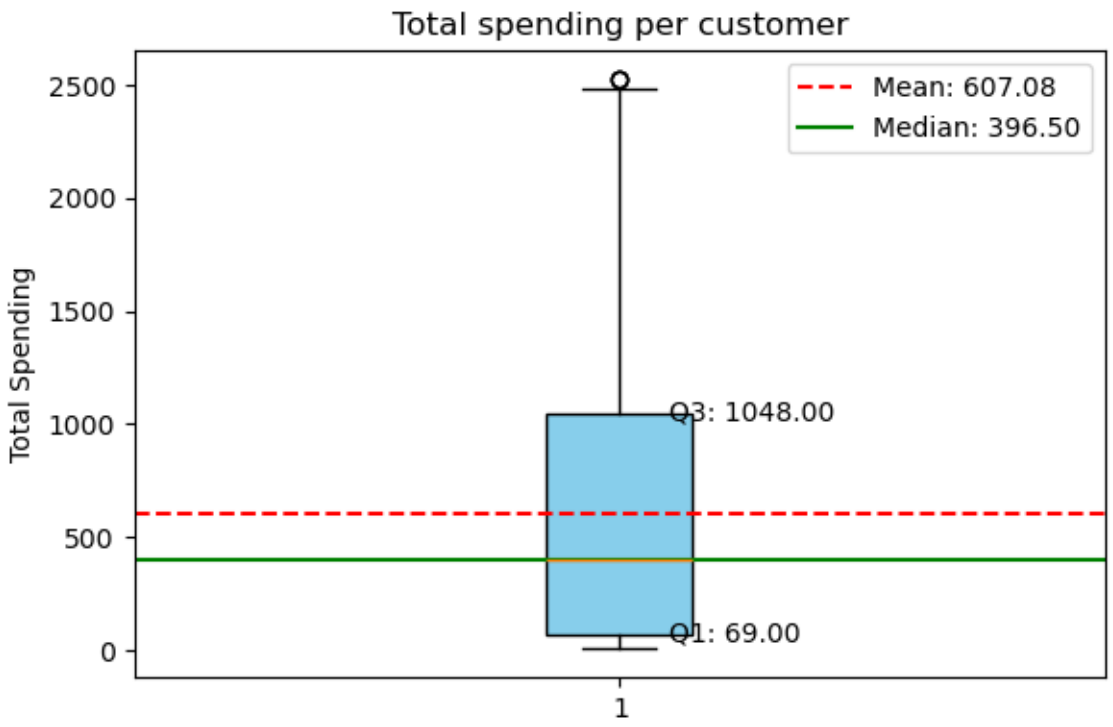
The most common education level is Graduation, representing 1116 customers (50.36%).

Marital status

- The majority of customers are either, Married (38.67%), Together (25.86%) or

Single (21.25%).

Higher spenders



count	2216
mean	607
std	603
min	5
25%	69
50%	397
75%	1048
max	2525

High value customers

Higher spenders: total spending>1048

- 553 customers
-

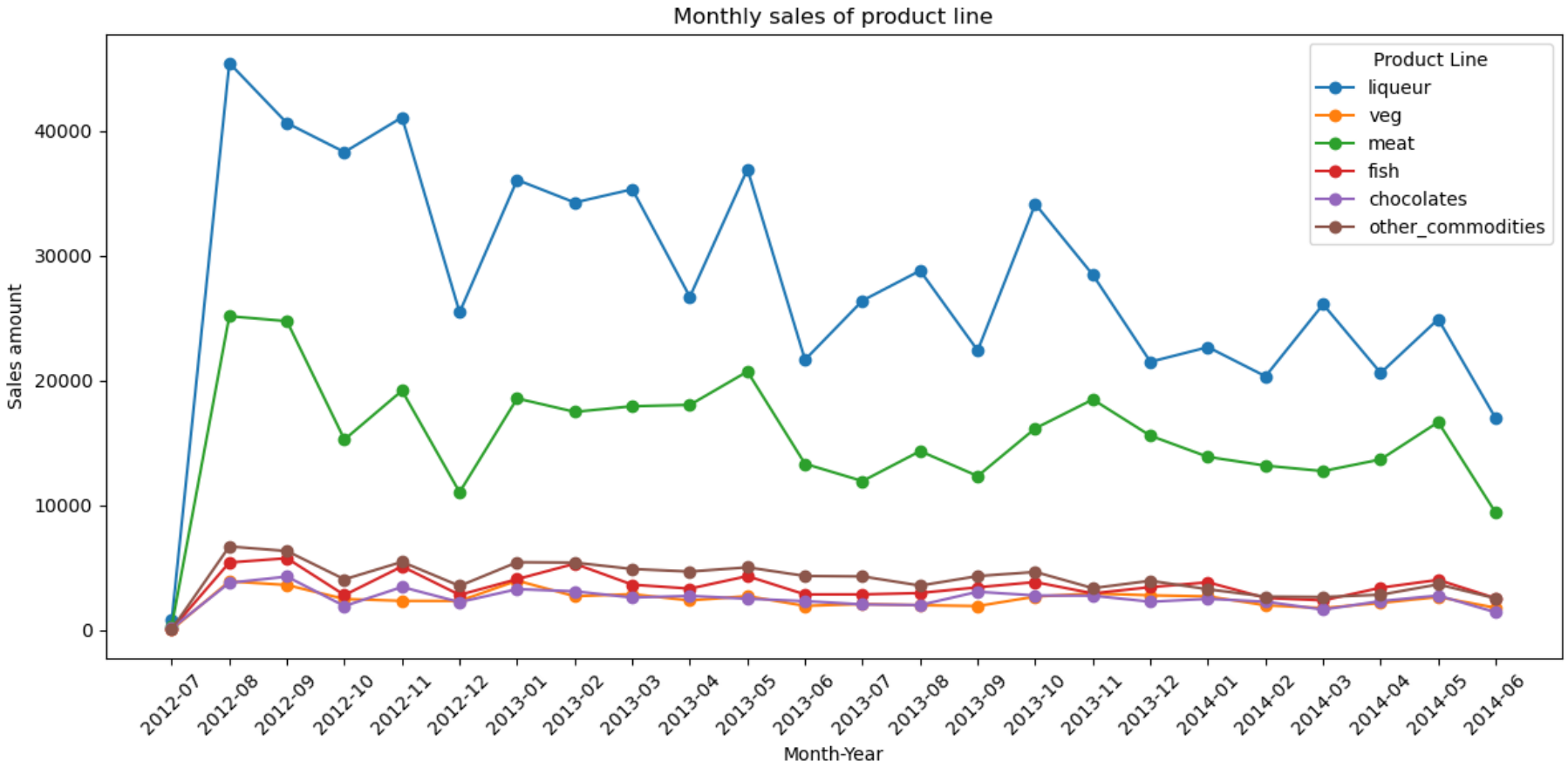
50-59	146
60-69	133
40-49	103
70-79	97
29-39	61
80+	13

Upper-Mid	380
Mid	159
High	7
Lower-Mid	6
Low	1

Spain	271
South Africa	89
Canada	70
Australia	34
India	33
Germany	31
United State	24
Montenegro	1

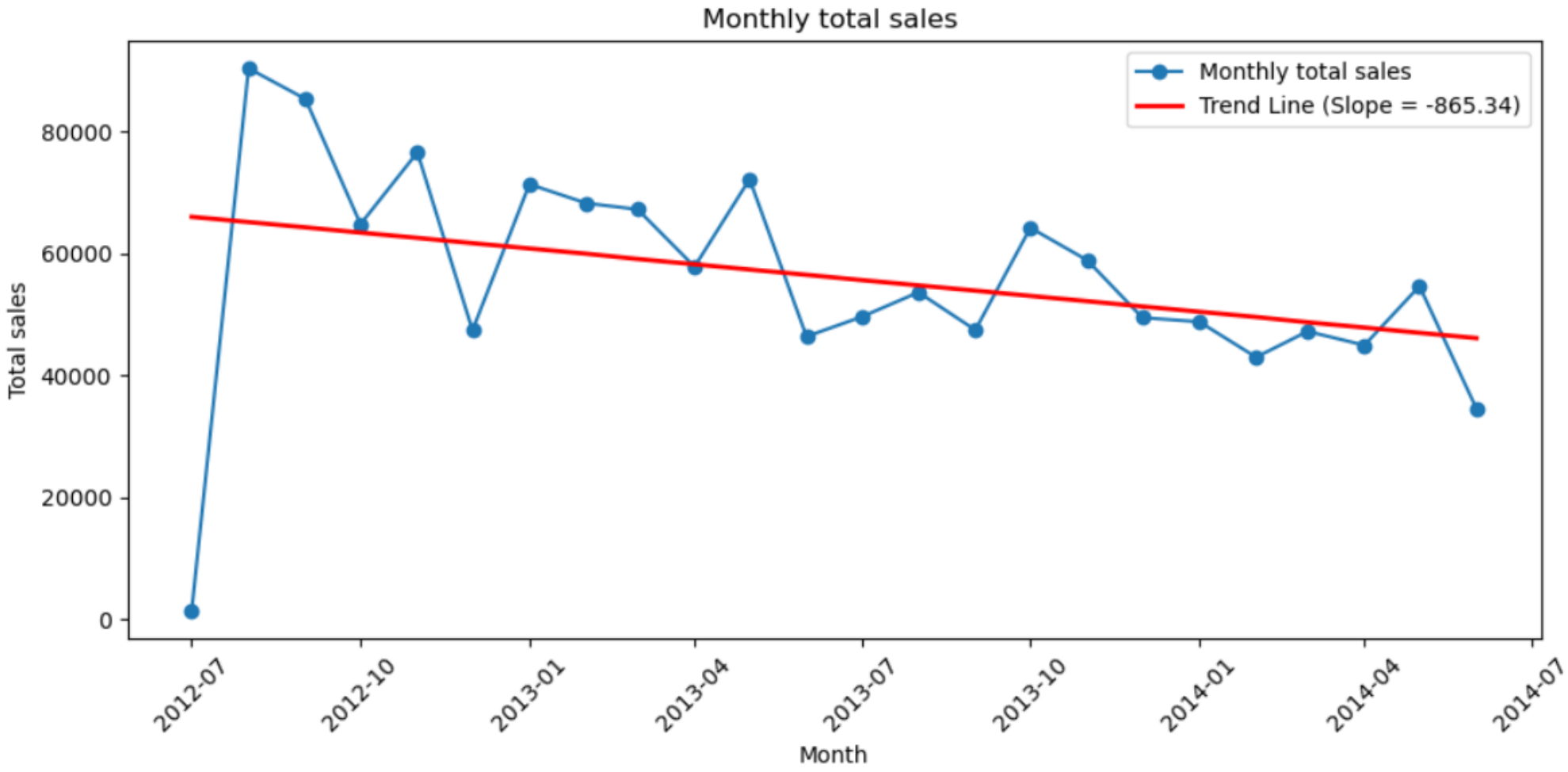
- Most higher spenders tend to be middle-aged, 40-69 yrs.
- They are predominantly in the upper-middle income group and are largely concentrated in Spain.
- Smaller number of higher spenders are coming from other countries.

Monthly sales



**** Monthly sales are on the decline**

Percentage Change from 2012-08 to 2014-06



slope=-865.34

On average, sales are declining by -865 units per month over the analysed period.

Effectiveness of advertising

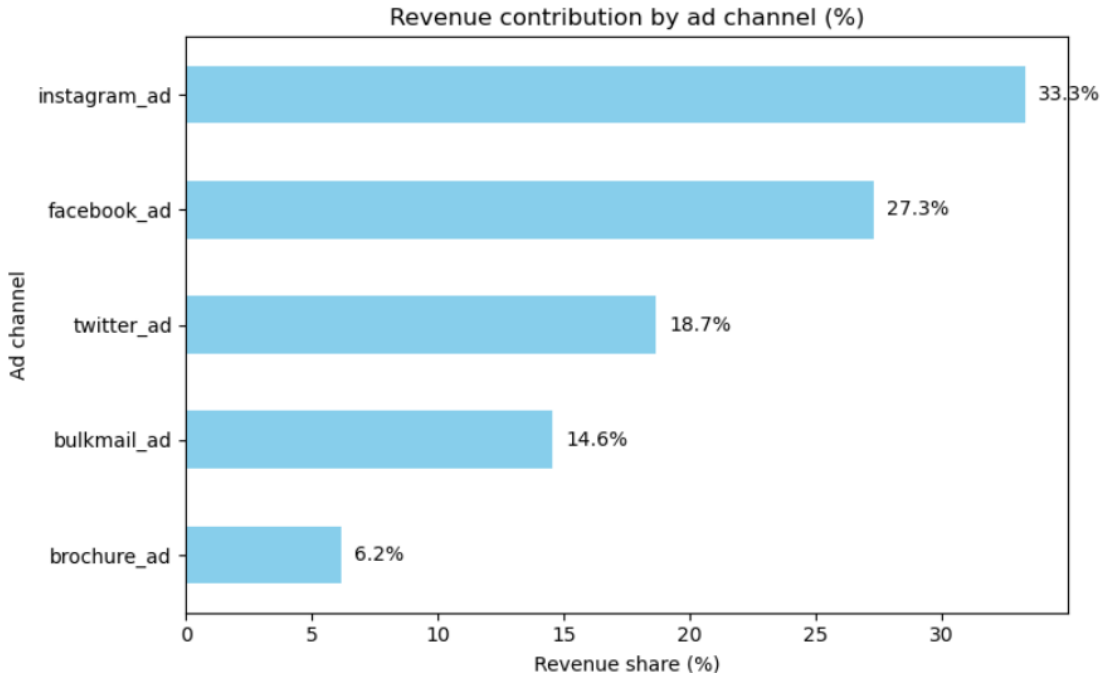
Instagram_ad	4.11%
Facebook_ad	3.56%
Bulkmail_ad	3.47%
Twitter_ad	2.80%
Brochure_ad	0.90%

- The ad campaign response rate: 15%
- The majority did not respond, suggesting room for improving target advertising.
- Digital channels, Instagram and Facebook generated higher engagement compared to traditional or less-targeted channels like brochures.

Responsiveness to ads

- Age: middle-aged customers, 40-69 yrs,11.1%
- Marital status: single, 4.7% and married, 4,2%
- Education level: graduates, 6.9% and PhDs, 4.6%
- Income: middle-to-upper income groups, 13%
- The majority from Spain, 7.9%

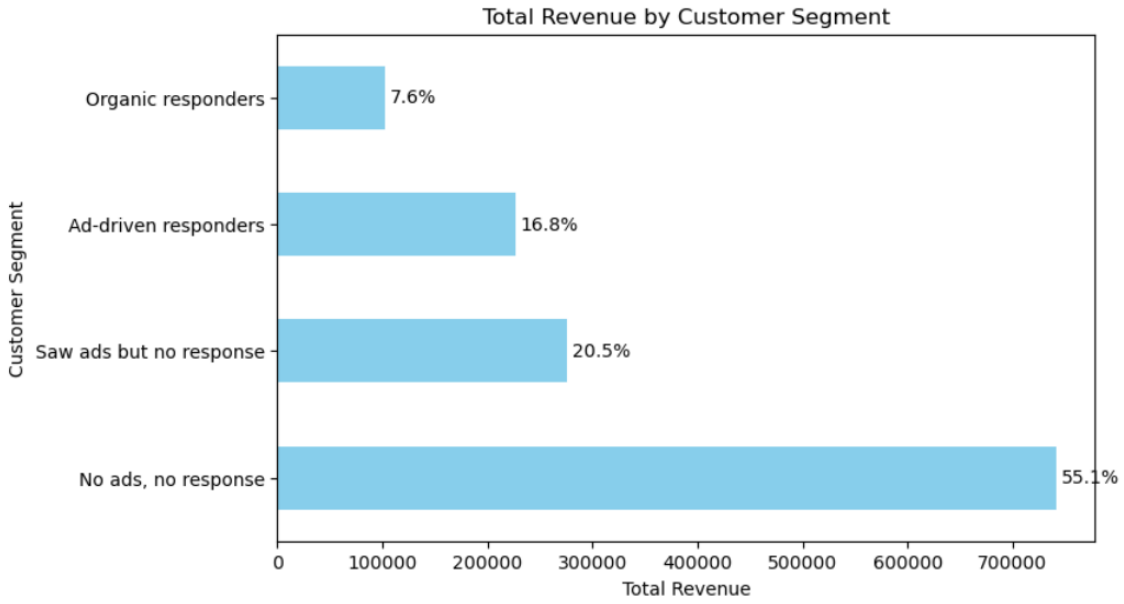
Ad channel effectiveness



Ad_ch	Revenue	%
instagram_ad	149664	33.3
facebook_ad	122636	27.3
twitter_ad	83805	18.7
bulkmail_ad	65391	14.6
brochure_ad	27657	6.16
Total	449,153	

Segmentation by response to ad campaign

Ads seen	Response	Customer type	Revenue	%
ch_exposure>0	1	Ad_driven responders	226341	16.8
ch_exposure=0	1	Organic responders	101884	7.6
ch_exposure>0	0	saw ads but did not respond	275681	20.5
ch_exposure=0	0	No ads and no response	741373	55.1
		Total	1345279	



Success of the ad campaign

Customers who saw ads and responded generated **much higher revenue (£226,341)** than organic responders (£101,884), which indicates that ads are effective at driving conversions.

However, a portion of customers saw ads but did not respond (20.5%) and a large group wasn't reached at all (55.1%).

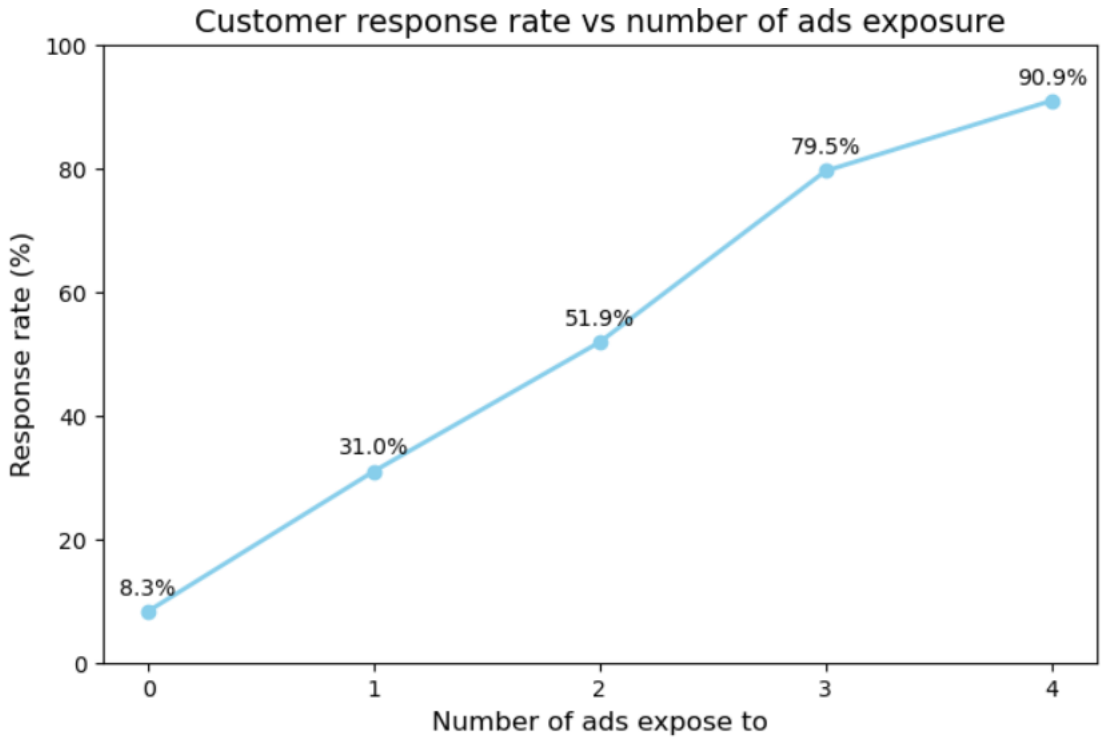
Ads do work, but not for everyone. Their effectiveness depends on targeting and reach.

Instagram ads generated the highest revenue in comparison to other channels.

Revenue generated by each customer segments

- 👉 **No customer in the dataset saw all 5 ads.**

channels_exposed_count	count	sum	response_rate %
0	1757	146	8.3
1	323	100	31.0
2	81	42	51.9
3	44	35	79.5
4	11	10	90.9



Revenue generated by each customer segments

Customers who were not exposed to any ads had a low response rate of about 8.3% (146 out of 1757).

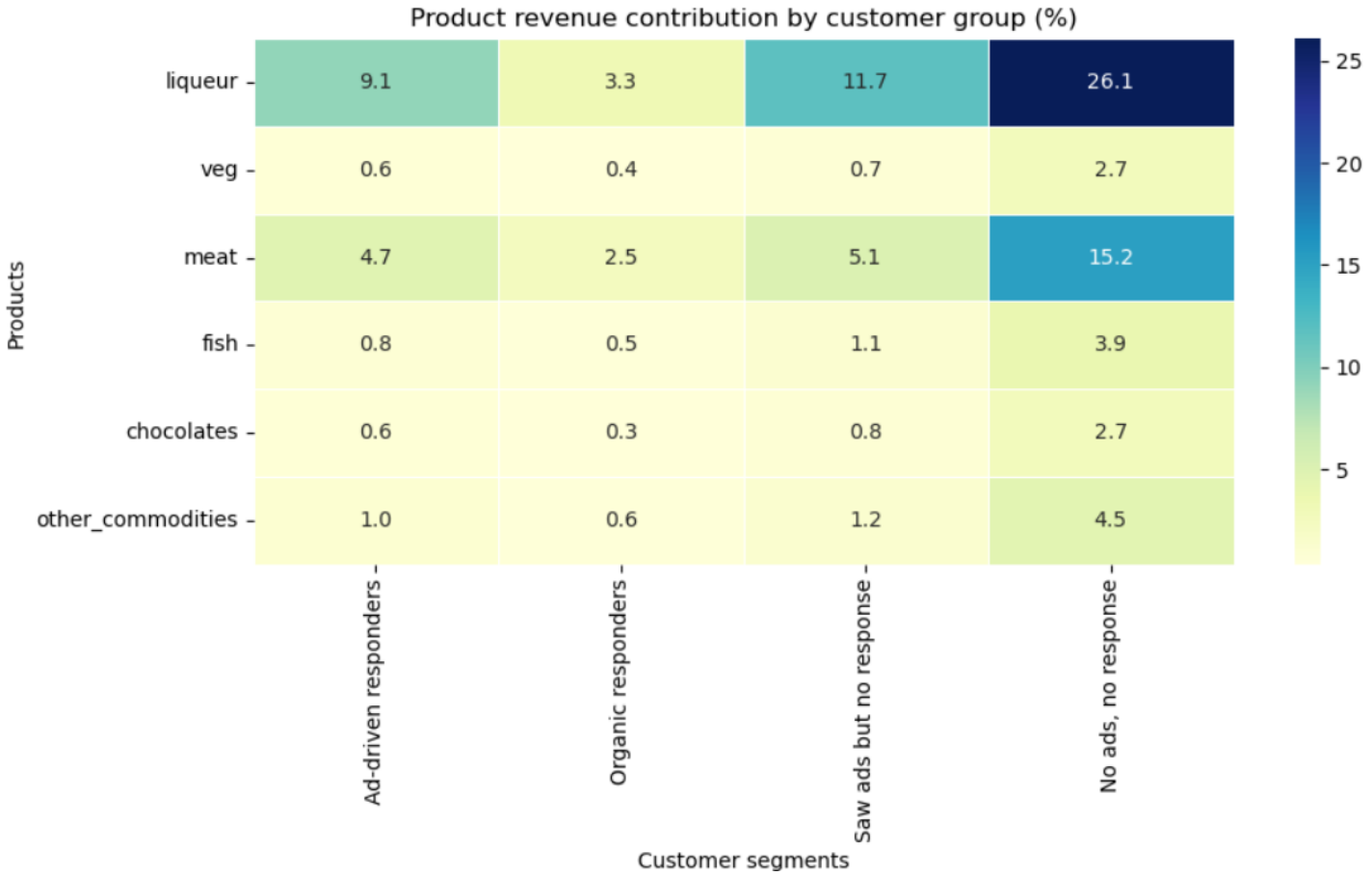
As exposure increases, both the number and proportion of positive responses rise significantly:

The trend becomes even stronger at higher exposure.

Customers who saw 3 ads had a response rate of about 79.5% and those who saw 4 ads reached over 90% .

There is a positive relationship between ad exposure and customer response, indicating that increased marketing efforts improve engagement. However, results for higher exposure levels should be interpreted with caution due to smaller sample sizes.

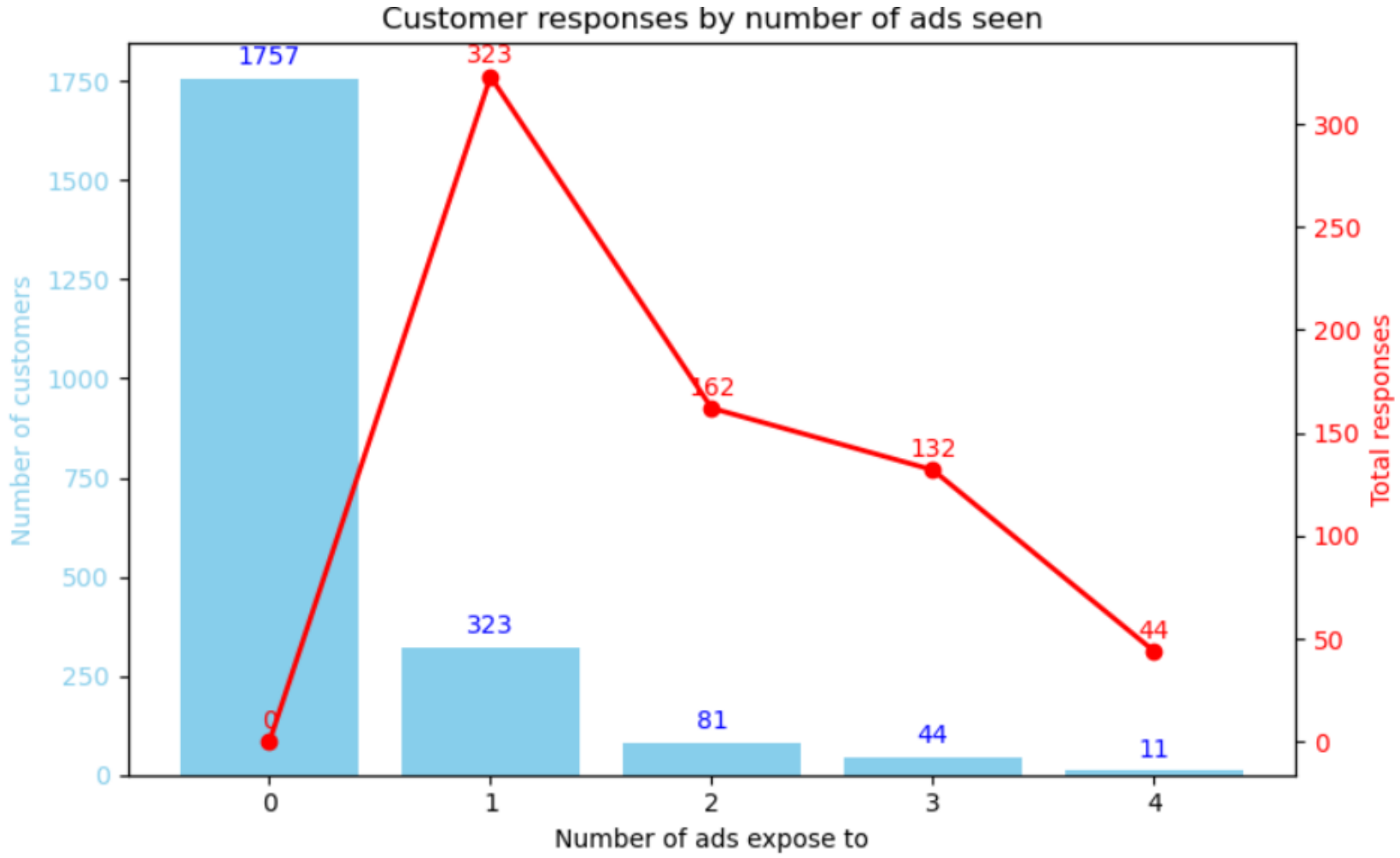
Revenue: product wise and customer segments



Revenue: product wise and customer segments

- Advertising influenced spending across product categories, particularly higher value items.
- All customers consistently spend more on liqueur and meat, suggesting these products are more influenced by advertising.
- Customers who saw ads but did not respond show relatively higher spending than those with no ad exposure, indicating some indirect impact of ads.
- Lower-value categories like veg, fish, and chocolates show smaller differences, implying they are less affected by advertising.
- Ads appear to be more effective in driving purchases of premium products.

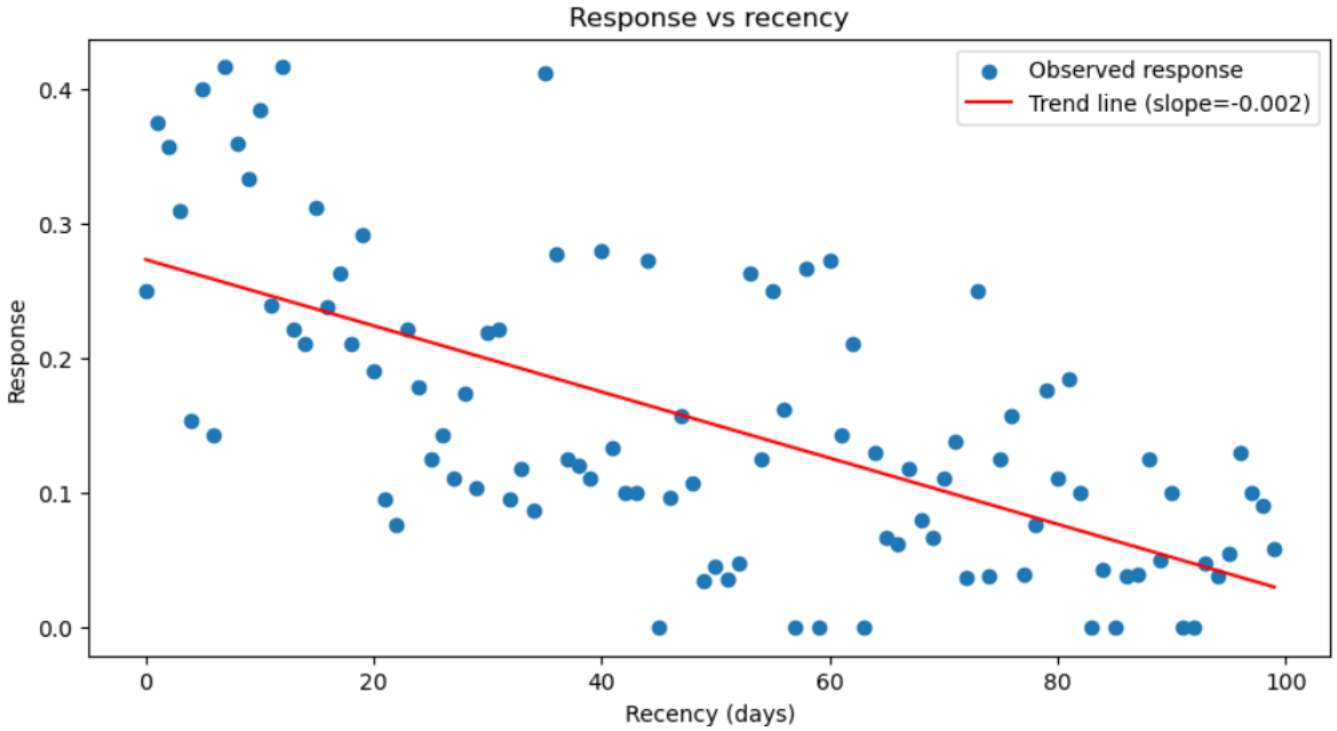
Customer exposure vs response to ads



Customer exposure to ads

- There is a clear pattern in how ad exposure affects customer behaviour.
- A large majority of customers saw no ads and generated no responses.
- A smaller segment of customers(20.67%) was exposed to ads.
- Among those who saw ads, engagement was extremely high.
- This highlights a key trade-off in marketing strategy, higher ad exposure results in lower engagement. Multiple ad exposures tend to reach fewer customers, but these customers are highly engaged.
- Single ad exposure is highly effective at driving initial engagement, focusing on carefully selected customers with multiple ad exposures can significantly increase response intensity and overall campaign impact.

Recency and response



For every 1-day increase in recency, the response decreases by 0.2% (since slope is -0.002 in units of response per day).

Recency and response

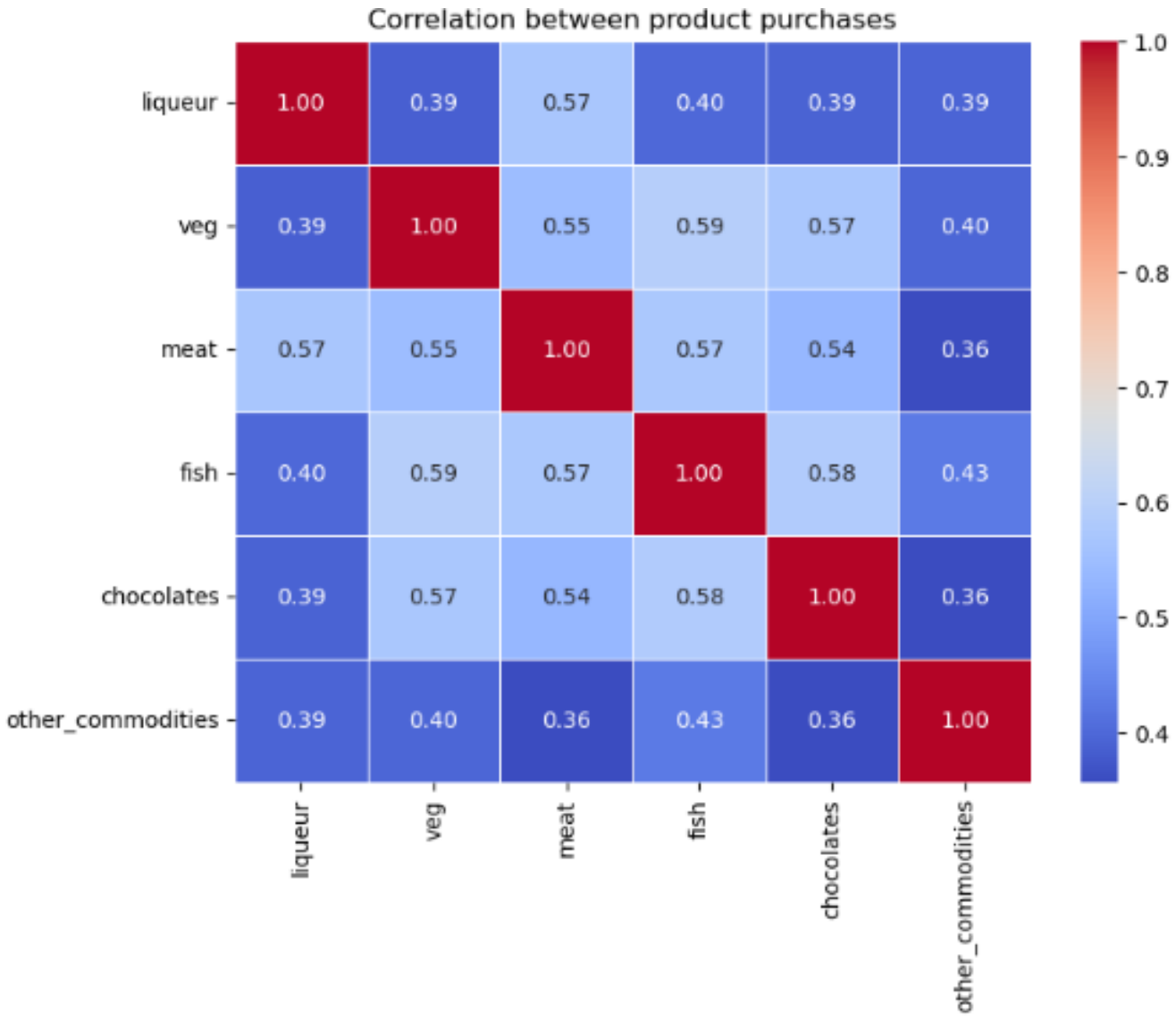
Correlation between recency and response: -0.200

The correlation is negative.

As recency increases (customer hasn’t purchased in a while) the likelihood of response decreases.

Customers who purchased more recently are slightly more likely to respond to a campaign than those who haven’t purchased in a while.

Customer purchase habits



Customer purchase habits

Liqueur and Meat (0.57) show a moderate positive correlation, suggesting these items are frequently purchased together.

Vegetables and Fish (0.59) have the strongest correlations, indicating customers buying vegetables often also purchase fish.

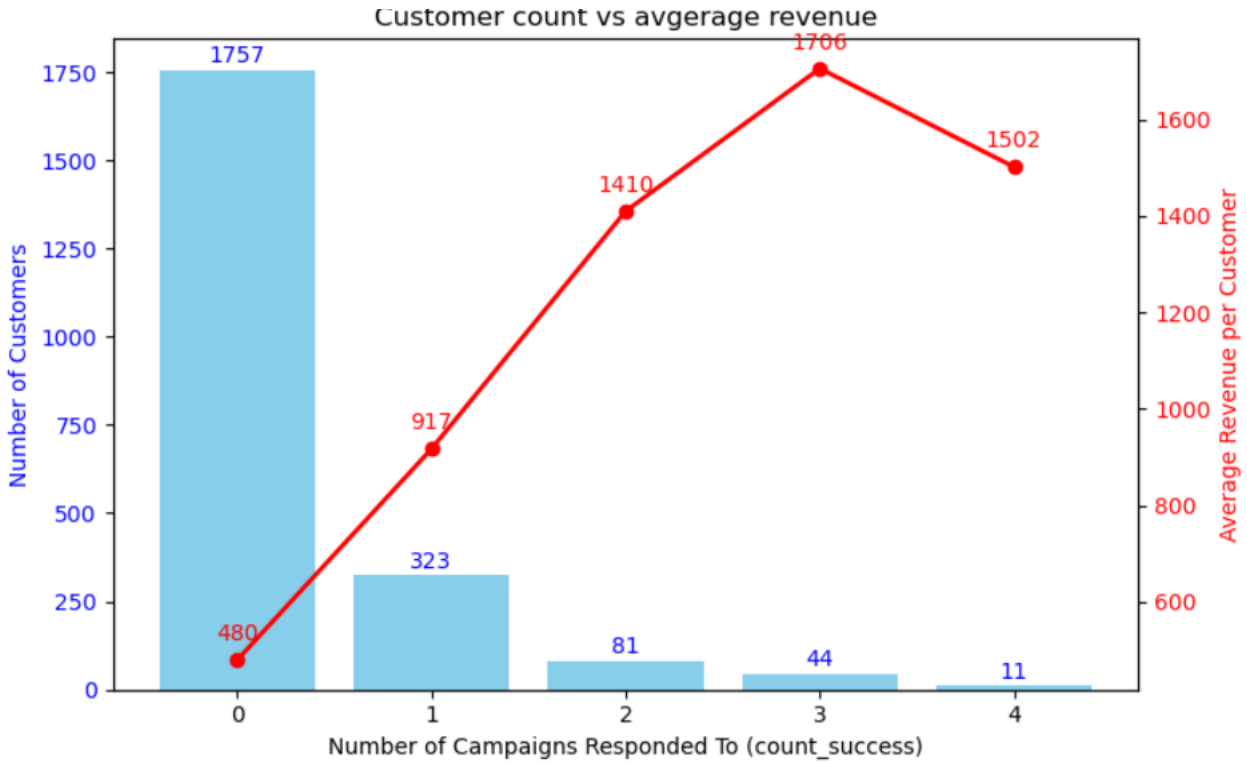
Fish and Chocolates (0.58) and Meat and Fish (0.57) also show moderate relationships, suggesting these categories are commonly bought by the same customers.

Other commodities show weaker correlations (around 0.35-0.43) with other products, indicating they are purchased more independently.

All correlations are positive, which suggests that customers who spend more generally tend to buy multiple products rather than only one type of product.

This pattern indicates opportunities for cross-selling and bundled promotions across product categories.

Customer numbers vs average revenue



Customer numbers vs average revenue

- Most customers (1757) did not respond to any campaign, and their average spending is the lowest.
- Customers who responded to one ad campaign spend almost twice as much on average compared with non-responders.
- Customers who responded to two or three campaigns show the highest average spending, indicating strong engagement with the brand.
- A very small group of customers responded to four campaigns; although they are few, they generate very high spending per customer.
- There is a strong positive relationship between campaign response and customer spending. Customers who engage with marketing campaigns tend to be significantly more valuable, suggesting that identifying and targeting responsive customers could be an effective strategy for increasing revenue.
- Customers who respond to multiple marketing campaigns demonstrate substantially higher average spending, indicating that campaign engagement is strongly associated with higher customer value.

Campaign Success

country	numwebbuys	response	count_success
Australia	4.05	0.15	0.23
Canada	4.29	0.14	0.33
Germany	4.00	0.15	0.33
India	3.97	0.09	0.26
Montenegro	6.00	0.67*	0.33
South Africa	4.15	0.15	0.26
Spain	4.00	0.16	0.32
United State	4.48	0.12	0.24

income_groups	count_success
Low	0.07
Lower-Mid	0.12
Mid	0.22
Upper-Mid	0.78
High	0.92

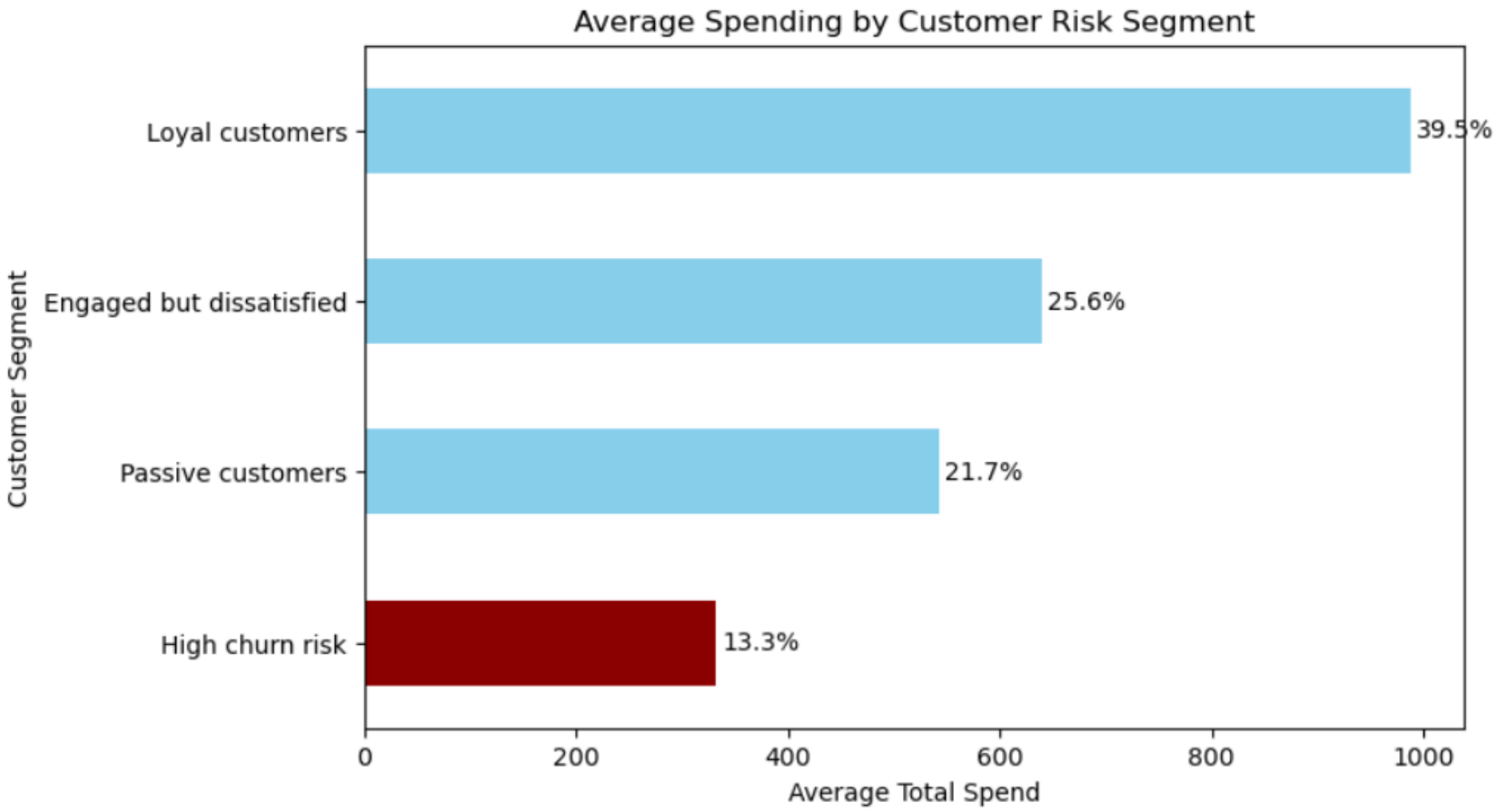
As income groups increase from Low to High, the campaign success rises significantly.

*fewer customers

Campaign Success

- A moderate variation in customer behaviour across countries.
- Most countries have similar web purchase levels, but response rates differ more noticeably.
- Spain and South Africa show relatively higher response rates, while India and the United States are lower.
- Spain, Canada and Germany stand out with higher campaign success count, suggesting effective targeting.
- Montenegro shows very high engagement across all metrics, but this is likely due to a small sample size.
- Purchasing behaviour is fairly consistent globally, campaign effectiveness varies by country, highlighting opportunities for more localised marketing strategies.

Risk analysis of customer churn



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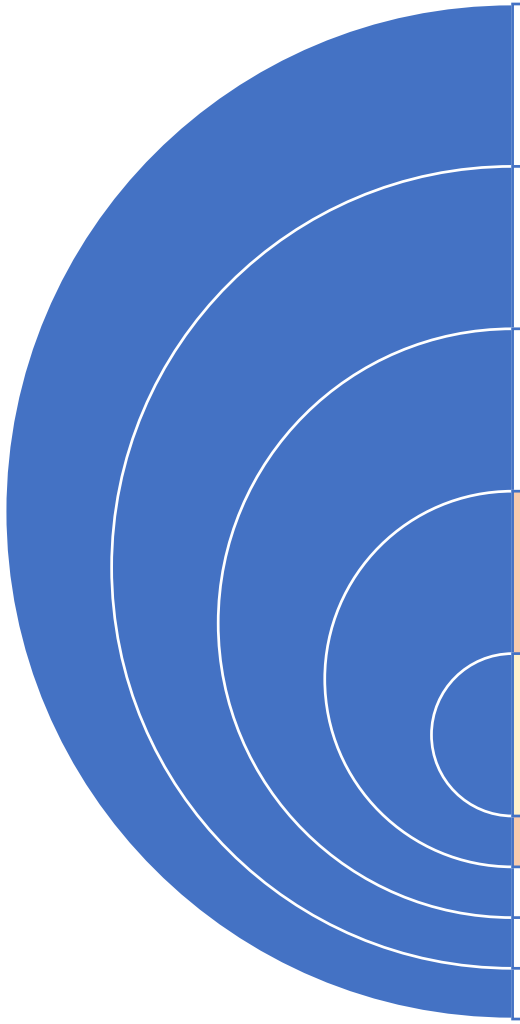

Passive customers dominate the customer base: represent the largest segment, contributing the highest total revenue of about 1.01 million.

However, these customers do not respond to campaigns, suggesting a large opportunity for better engagement strategies.

Loyal customers generate high value per customer. Although there are only 330 loyal customers, they contribute over 326k in revenue, indicating strong spending behaviour and engagement with marketing campaigns.

High churn risk customers are very small but concerning. Only 18 customers fall into this category, contributing about 5,986 in revenue. These customers have complained and do not respond to campaigns, suggesting dissatisfaction and a high likelihood of leaving.

Engaged but dissatisfied customers are rare. There are only 3 customers in this segment, contributing about 1,919 in revenue. Despite complaining, they respond to campaigns, indicating that addressing their concerns could help retain them.

Conclusion

Revenue decline is primarily due to low engagement among passive customers.

Digital campaigns are effective for a smaller, high-value segment, but the majority remain unresponsive.

Ad exposure and targeted marketing strongly correlate with increased spending, especially for premium products and high-income, educated, or middle-aged customers.

Strategic Recommendations

- Focus marketing on high-value, responsive segments for greater ROI.
- Personalise campaigns using customer data to improve engagement.
- Optimise ad frequency to balance reach and intensity.
- Explore cross-selling opportunities based on product correlations.
- Address concerns of dissatisfied or high-churn customers to prevent revenue loss.